Taller 2-2: Catálogo

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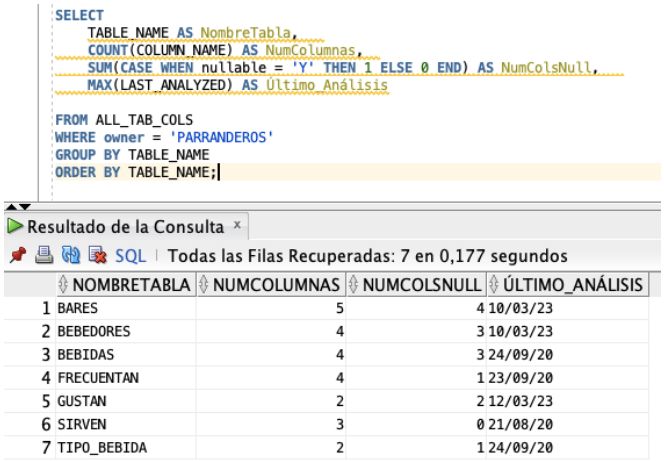
1 Consulta 1

```
SELECT
    TABLE_NAME AS NombreTabla,
    COUNT(COLUMN_NAME) AS NumColumnas,
    SUM(CASE WHEN nullable = 'Y' THEN 1 ELSE 0 END) AS NumColsNull,
    MAX(LAST_ANALYZED) AS Último_Análisis

FROM ALL_TAB_COLS
WHERE owner = 'PARRANDEROS'
GROUP BY TABLE_NAME
ORDER BY TABLE_NAME;
```



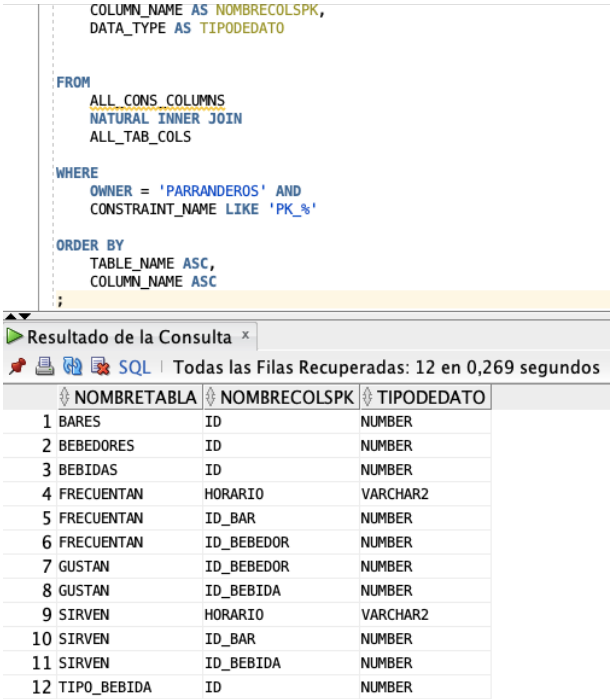
	NOMBRETABLA	NUMCOLUMNAS	NUMCOLSNULL	ÚLTIMO_ANÁLISIS
1	BARES	5	4	10/03/23
2	BEBEDORES	4	3	10/03/23
3	BEBIDAS	4	3	24/09/20
4	FRECUENTAN	4	1	23/09/20
5	GUSTAN	2	2	12/03/23
6	SIRVEN	3	0	21/08/20
7	TIPO_BEBIDA	2	1	24/09/20

Figura 1. Resultados de la consulta 1.

2 Consulta 2

```
SELECT
    TABLE_NAME AS NOMBRETABLA,
```

```
    COLUMN_NAME AS NOMBRECOLSPK,
    DATA_TYPE AS TIPODEDATO

FROM
    ALL_CONS_COLUMNS
    NATURAL INNER JOIN
    ALL_TAB_COLS

WHERE
    OWNER = 'PARRANDEROS' AND
    CONSTRAINT_NAME LIKE 'PK_%'

ORDER BY
    TABLE_NAME ASC,
    COLUMN_NAME ASC;
```



```
    COLUMN_NAME AS NOMBRECOLSPK,
    DATA_TYPE AS TIPODEDATO

FROM
    ALL_CONS_COLUMNS
    NATURAL INNER JOIN
    ALL_TAB_COLS

WHERE
    OWNER = 'PARRANDEROS' AND
    CONSTRAINT_NAME LIKE 'PK_%'

ORDER BY
    TABLE_NAME ASC,
    COLUMN_NAME ASC
;
```

Resultado de la Consulta

SQL | Todas las Filas Recuperadas: 12 en 0,269 segundos

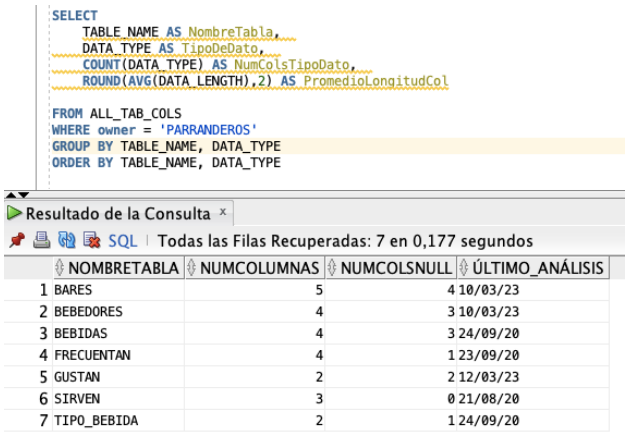
	NOMBRETABLA	NOMBRECOLSPK	TIPODEDATO
1	BARES	ID	NUMBER
2	BEBEDORES	ID	NUMBER
3	BEBIDAS	ID	NUMBER
4	FRECUENTAN	HORARIO	VARCHAR2
5	FRECUENTAN	ID_BAR	NUMBER
6	FRECUENTAN	ID_BEBEDOR	NUMBER
7	GUSTAN	ID_BEBEDOR	NUMBER
8	GUSTAN	ID_BEBIDA	NUMBER
9	SIRVEN	HORARIO	VARCHAR2
10	SIRVEN	ID_BAR	NUMBER
11	SIRVEN	ID_BEBIDA	NUMBER
12	TIPO_BEBIDA	ID	NUMBER

Figura 2. Resultados de la consulta 2.

3 Consulta 3

```
SELECT
    TABLE_NAME AS NombreTabla,
    DATA_TYPE AS TipoDeDato,
    COUNT(DATA_TYPE) AS NumColsTipoDato,
    ROUND(AVG(DATA_LENGTH),2) AS PromedioLongitudCol

FROM ALL_TAB_COLS
WHERE owner = 'PARRANDEROS'
```

```
GROUP BY TABLE_NAME, DATA_TYPE
ORDER BY TABLE_NAME, DATA_TYPE
```



```
SELECT
    TABLE_NAME AS NombreTabla,
    DATA_TYPE AS TipoDeDato,
    COUNT(DATA_TYPE) AS NumColsTipoDato,
    ROUND(AVG(DATA_LENGTH),2) AS PromedioLongitudCol

FROM ALL_TAB_COLS
WHERE owner = 'PARRANDEROS'
GROUP BY TABLE_NAME, DATA_TYPE
ORDER BY TABLE_NAME, DATA_TYPE
```

Resultado de la Consulta

SQL | Todas las Filas Recuperadas: 7 en 0,177 segundos

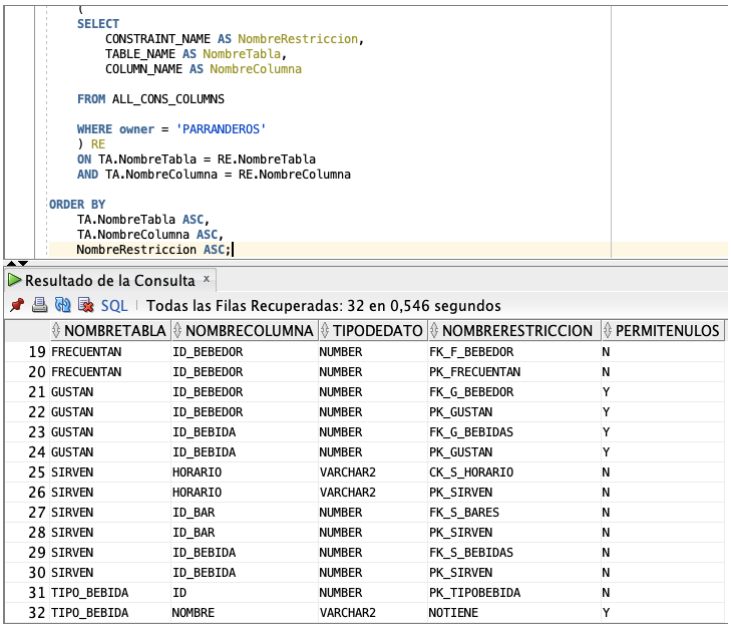
	NOMBRETABLA	NUMCOLUMNAS	NUMCOLSNULL	ÚLTIMO_ANÁLISIS
1	BARES	5	4	10/03/23
2	BEBEDORES	4	3	10/03/23
3	BEBIDAS	4	3	24/09/20
4	FRECUENTAN	4	1	23/09/20
5	GUSTAN	2	2	12/03/23
6	SIRVEN	3	0	21/08/20
7	TIPO_BEBIDA	2	1	24/09/20

Figura 3. Resultados de la consulta 3.

4 Consulta 4

```
SELECT
    TA.NombreTabla,
    TA.NombreColumna,
    TA.TipoDeDato,
    NVL(RE.NombreRestriccion, 'NOTIENE') AS NombreRestriccion,
    TA.PermiteNulos

FROM
    (
    SELECT
        TABLE_NAME AS NombreTabla,
        COLUMN_NAME as NombreColumna,
        DATA_TYPE AS TipoDeDato,
        NULLABLE AS PermiteNulos

    FROM ALL_TAB_COLS

    WHERE owner = 'PARRANDEROS'

    GROUP BY
        TABLE_NAME,
        COLUMN_NAME,
        DATA_TYPE,
        NULLABLE

    ORDER BY
        TABLE_NAME,
        COLUMN_NAME,
```

```
        DATA_TYPE,
        NULLABLE
    ) TA
    LEFT JOIN
    (
    SELECT
        CONSTRAINT_NAME AS NombreRestriccion,
        TABLE_NAME AS NombreTabla,
        COLUMN_NAME AS NombreColumna

    FROM ALL_CONS_COLUMNS

    WHERE owner = 'PARRANDEROS'
    ) RE
    ON TA.NombreTabla = RE.NombreTabla
    AND TA.NombreColumna = RE.NombreColumna

ORDER BY
    TA.NombreTabla ASC,
    TA.NombreColumna ASC,
    NombreRestriccion ASC;
```



Resultado de la Consulta

Todas las Filas Recuperadas: 32 en 0,546 segundos

	NOMBRETABLA	NOMBRECOLUMNA	TIPODEDATO	NOMBRERESTRICCION	PERMITENULOS
19	FRECUENTAN	ID_BEBEDOR	NUMBER	FK_F_BEBEDOR	N
20	FRECUENTAN	ID_BEBEDOR	NUMBER	PK_FRECUENTAN	N
21	GUSTAN	ID_BEBEDOR	NUMBER	FK_G_BEBEDOR	Y
22	GUSTAN	ID_BEBEDOR	NUMBER	PK_GUSTAN	Y
23	GUSTAN	ID_BEBIDA	NUMBER	FK_G_BEBIDAS	Y
24	GUSTAN	ID_BEBIDA	NUMBER	PK_GUSTAN	Y
25	SIRVEN	HORARIO	VARCHAR2	CK_S_HORARIO	N
26	SIRVEN	HORARIO	VARCHAR2	PK_SIRVEN	N
27	SIRVEN	ID_BAR	NUMBER	FK_S_BARES	N
28	SIRVEN	ID_BAR	NUMBER	PK_SIRVEN	N
29	SIRVEN	ID_BEBIDA	NUMBER	FK_S_BEBIDAS	N
30	SIRVEN	ID_BEBIDA	NUMBER	PK_SIRVEN	N
31	TIPO_BEBIDA	ID	NUMBER	PK_TIPOBEBIDA	N
32	TIPO_BEBIDA	NOMBRE	VARCHAR2	NOTIENE	Y

Figura 4. Resultados de la consulta 4.

5 Consulta 5

Sentencia de la primera parte:

```
SELECT
    'SELECT TABLE_NAME, COLUMN_NAME, COUNT(CONSTRAINT_NAME) ' ||
    'FROM ALL_CONS_COLUMNS ' ||
```

```
    'WHERE OWNER = ''PARRANDEROS'' AND ' ||
    'TABLE_NAME = ''' || TABLE_NAME ||''' AND ' ||
    'COLUMN_NAME = ''' || COLUMN_NAME || ''' ' ||
    'GROUP BY TABLE_NAME, COLUMN_NAME;'

FROM
    ALL_TAB_COLS

WHERE
    OWNER = 'PARRANDEROS' AND
    COLUMN_NAME NOT LIKE '%R%' AND
    REGEXP_COUNT(COLUMN_NAME, '[AEIOUaeiou]') <= 2;
```

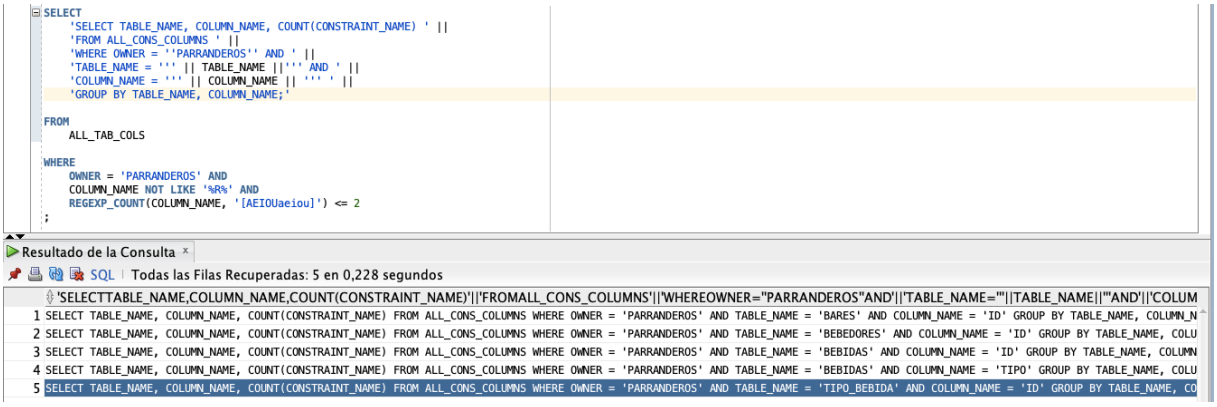


Figura 5. Resultados de la ejecución de la primera fase de la consulta 5.

Scripts generados para la segunda parte:

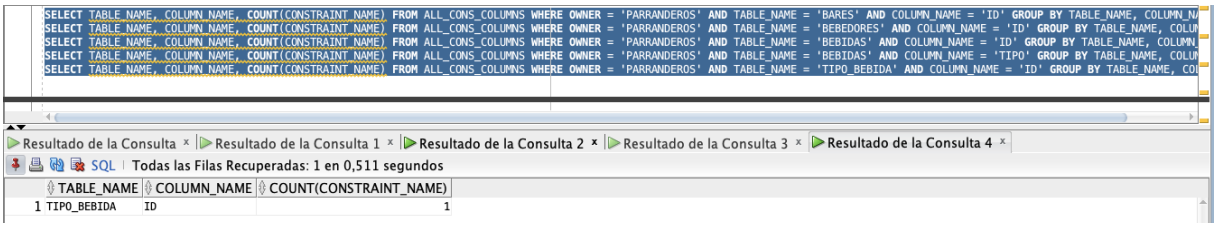


Figura 6. Resultados de la ejecución de los *scripts*.